

Steffi Roy

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EDUCATION

Northeastern University

Boston, MA | Jan 2026 – Present

M.S. in Computer Science (Khoury College) · GPA 4.0 / 4.0

Key Courses: Foundations of AI, Human-Computer Interaction, Algorithms, Pattern Recognition and Computer Vision (ongoing)

University of Florida

Gainesville, FL | Jan 2021 – Aug 2023

M.S. in Electrical & Computer Engineering · GPA 3.6 / 4.0

Key Courses: Hardware Security, Computer Architecture, Machine Learning, Quantum Computing

Mahindra Ecole Centrale

Hyderabad, India | Jul 2016 – Aug 2020

B.Tech in Electrical & Electronics Engineering · CGPA 9.3 / 10

RESEARCH EXPERIENCE

University of Florida - FICS Lab, Research Assistant

Dec 2021 – Jan 2023

- ▶ Developed HWGN2, a hardware-garbled neural network framework providing side-channel resilience for secure deep learning inference on FPGA; published and presented at *SPACE 2022*.
- ▶ Designed privacy-preserving EDA using garbled circuits and multiparty computation (SFE/PFE) to protect IP and PDKs; published at *ICCAD 2022*.
- ▶ Implemented IC metering protocols using Bellare-Micali & Naor-Pinkas oblivious transfer with countermeasures against fault injection; published and presented at *SRC TECHCON 2022*.

PROJECTS

Multi-Agent Reinforcement Learning for Physical Deception Environment 🏠 📖 Northeastern University

- ▶ Evaluated Multi Agent Deep Deterministic Policy Gradient (MADDPG) against Independent DDPG in physical deception environment. Extended PettingZoo library for scaled adversary configuration; analyzed strategic reasoning and communication under partial observability.
- ▶ Quantified performance via reward convergence and adversary reach rate over 10k episodes, identifying CTDE advantages for misleading adversaries in mixed cooperative-competitive environments.

LLM Compression Efficient Training 🏠 SUNDAL Hackathon

- ▶ Optimized a transformer model for the OpenAI Parameter Golf challenge, achieving a strict 16MB artifact size by implementing Quantization-Aware Training (QAT) with int6 precision and straight-through estimator (STE) gradients. Utilized the Muon second-order optimizer to accelerate convergence, reaching bits-per-byte loss on the FineWeb dataset within a 10-minute training window on 8xH100 GPUs.
- ▶ Integrated score-first Test-Time Training (TTT) to enable the model to dynamically adapt to validation sequences at inference time, significantly improving performance without exceeding memory constraints.

HabitBuddy, HCI Application Design 🏠 📖 Northeastern University

- ▶ Led user research, paper prototyping, heuristic evaluation, and high-fidelity Figma prototyping for a companion-based habit tracker; iterative findings drove all key interaction redesign decisions.

PROFESSIONAL EXPERIENCE

Texas Instruments | Design Verification Engineer

Aug 2023 – Present

- ▶ SoC-level formal and simulation verification (Flash, XSPI, PLL, RTC); built Python/Tcl automation tooling; presented *Maestro XML Diff* tool at *DVCON 2025*.

Microsoft | Silicon Engineer Intern

Jan 2023 – Apr 2023

- ▶ Conducted full-chip AXI interconnect tests for Xbox platform; code coverage analysis and debugged code for SoC-to-Southbridge communication.

TECHNICAL SKILLS

Languages Python, C++, SystemVerilog / Verilog, Assembly, Tcl, SQL

ML / AI PyTorch, NumPy, OpenAI Gym, Stable-Baselines3, PEFT

Tools JasperGold, GEM5, Vivado, MATLAB, ModelSim, Git, Linux, L^AT_EX